Classifying Cardiomegaly: The KNN Baseline

Mini-Project Documentation for Final Year Evaluation

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Branch: Computer Science and Engineering

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Project Background & Literature Review

Establishing foundations in medical image classification

Research Objectives



Diagnosis

Early detection of Cardiomegaly (CTR > 0.5) through automated screening of digital chest radiography.



Efficiency

Implementation of fully vectorized distance metrics to maximize computational throughput on unified memory.

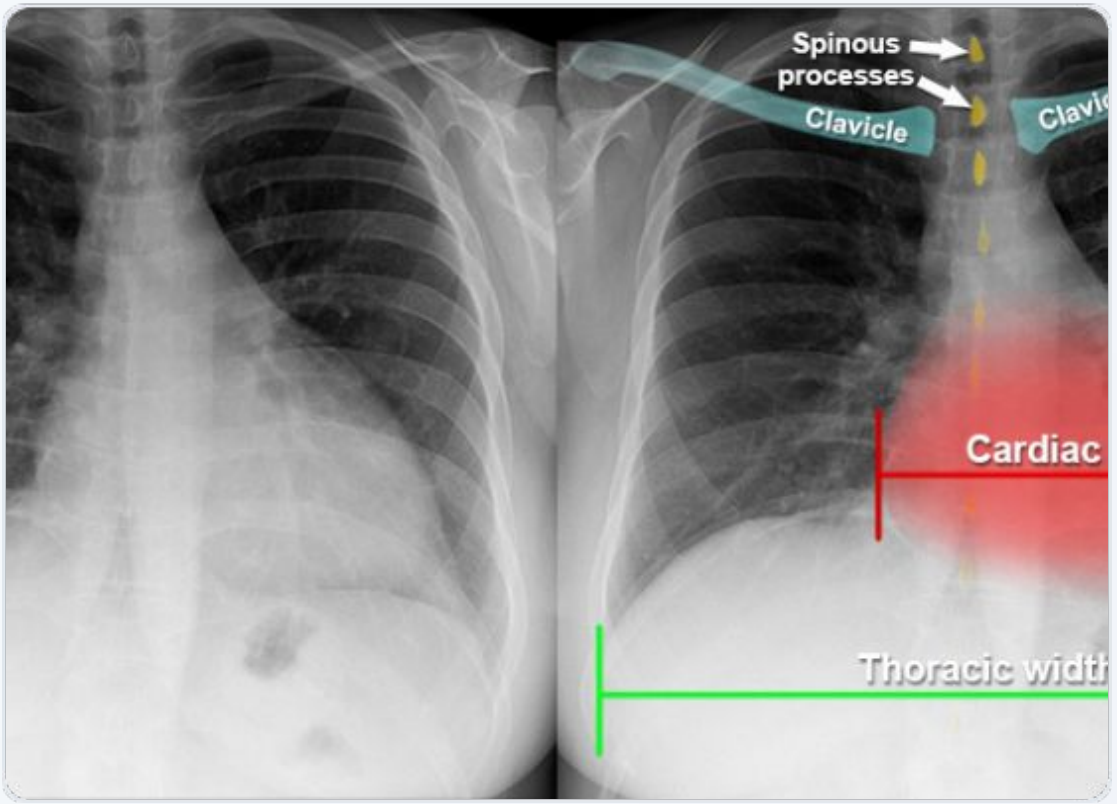


Calibration

Addressing clinical class imbalance in ChestMNIST to prevent model collapse toward majority-class bias.

Dataset: ChestMNIST (MedMNIST v2)

- **Scale:** 112,180 total images; educational subset of NIH ChestX-ray14.
- **Resolution:** Pre-processed 28x28 grayscale matrices (784 features).
- **Target:** Label Index 1 (Cardiomegaly) isolated for binary classification.
- **Imbalance:** Severe skew (~97.4% Negative) requiring undersampling.



Methodology: Vectorized KNN

L2 Distance Optimization

To avoid $O(N \cdot M)$ loops, the Euclidean distance is expanded using the polynomial theorem for matrix broadcasting.

$$\sqrt{\sum (x - y)^2} = \sqrt{x^2 - 2xy + y^2}$$

0-Loop Implementation

- **x^2:** Squared test norms.
- **-2xy:** Matrix dot product of test and train transposed.
- **y^2:** Squared training norms.
- **Result:** Near-instant distance tensor calculation for 10k+ images.

The Accuracy Paradox

97.4%

Raw Naive Accuracy

The Fallacy of High Accuracy

A naive model achieved 97.4% accuracy by blindly predicting "Negative" for every patient, resulting in:

- **Zero True Positives:** Failed to detect a single disease case.
- **Zero Recall:** Clinically useless performance in a diagnostic setting.

Design, Implementation & Deployment

From algorithmic core to interactive dashboard

Deployment: Streamlit Dashboard



Interactive Model Interface

- **Dynamic k-Selection:** Users can tune neighbor counts and see real-time updates.
- **Live Metrics:** Instant calculation of Precision, Recall, and F1-score.
- **Visual Heatmaps:** Automated Confusion Matrix generation via Seaborn.

Final Balanced Model Metrics (k=5)



> *Clinical Insight:* Balancing the data forced the model to prioritize **Recall**. While False Positives increased, the model successfully transitioned from detecting 0 cases to **63% of all diseased patients**.

Comparative Performance Summary

Benchmark Metric	Naive Unbalanced KNN	Balanced Baseline (k=5)
True Positives (Recall)	0 (0%)	364 (63%)
False Alarms (Precision)	0	9,645
Raw Accuracy	**97.4%**	56.0%
Clinical Applicability	None (Biased)	Screening Ready

Impact & Algorithmic Insights

- **Spatial Awareness:** KNN lacks "translational invariance." Flattening 2D images to 1D vectors causes a loss of spatial heart-lung context, leading to high false positives.
- **Scalability:** Establishing this baseline provides a definitive metric that future CNN architectures (ResNet/DenseNet) must objectively outperform.
- **Societal Benefit:** Automated screening tools assist radiologists in high-load environments, particularly in low-resource clinics where early Cardiomegaly detection is critical.

Questions?

Thank you for your time and guidance.

GitHub: github.com/Tendor517/Classifying-Cardiomegaly-The-KNN-Baseline

Contact: tenzin.dorjee@btech.christuniversity.in

Image Sources

https://www.radiologymasterclass.co.uk/images/chest-images/gallery-cardiac/cardiomegaly_mild.jpg?mtime=20210304211314&focal=none

Source: www.radiologymasterclass.co.uk

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https://static.vecteezy.com/system/resources/thumbnails/081/034/672/small/ai-brain-dashboard-icon-set-machine-learning-ui-browser-interface-for-tech-commercial-use-vector.jpg

Source: www.vecteezy.com